



ML Lab Week 10 SVM Lab

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SECTION: F

Analysis Questions:

1. *Based on the metrics and the visualizations, what inferences about the performance of the Linear Kernel can you draw?*

The Linear Kernel struggles with the Moons dataset (data points are interlocking half-moons) where the decision boundary needs to be curved. This shows that linear models work best when the data naturally forms linear patterns.

2. *Compare the decision boundaries of the RBF and Polynomial kernels. Which one seems to capture the shape of the data more naturally?*

The RBF kernel captures the shape of the data more naturally. It creates smooth, flexible boundaries that adapt well to the data's curvature, while the Polynomial kernel is curved, it produces more rigid boundaries.

3. *In this case, (banknote) which kernel appears to be the most effective?*

Linear kernel is most effective because the dataset is largely linearly separable.

4. *The Polynomial kernel shows lower performance here compared to the Moons dataset. What might be the reason for this?*

The Banknote data is more linearly separable, so the Polynomial kernel's added complexity doesn't help, it's introducing unnecessary non-linearity.

5. *Compare the two plots. Which model, the "Soft Margin" ($C=0.1$) or the "Hard Margin" ($C=100$), produces a wider margin?*

The Soft Margin produces a wider margin. With a smaller C value, the model prioritizes maximizing the margin width over perfectly classifying every training point, resulting in a more generous separation zone.

6. *Look closely at the "Soft Margin" ($C=0.1$) plot. You'll notice some points are either inside the margin or on the wrong side of the decision boundary. Why does the SVM allow these "mistakes"? What is the primary goal of this model?*

The Soft Margin SVM allows some misclassifications because its primary goal is to find a decision boundary that generalizes well to new data. By tolerating a few errors on the training set, it avoids fitting too closely to outlier points, which helps prevent overfitting.

7. Which of these two models do you think is more likely to be overfitting to the training data? Explain your reasoning.

The Hard Margin is more likely to overfit. It tries to classify every training point correctly, which means it might make the decision boundary accommodate outliers, making it less flexible for unseen data.

8. Imagine you receive a new, unseen data point. Which model do you trust more to classify it correctly? Why? In a real-world scenario where data is often noisy, which value of C (low or high) would you generally prefer to start with?

I'd trust the Soft Margin more for new data. In real-world scenarios with noisy data, starting with a lower C value is generally preferable because it creates a more robust model that doesn't overfit every training point and generalizes better to unseen examples.

Screenshots:

Moons dataset:

SVM with LINEAR Kernel <PES2UG23CS387>				
	precision	recall	f1-score	support
0	0.85	0.89	0.87	75
1	0.89	0.84	0.86	75
accuracy			0.87	150
macro avg	0.87	0.87	0.87	150
weighted avg	0.87	0.87	0.87	150

SVM with RBF Kernel <PES2UG23CS387>

	precision	recall	f1-score	support
0	0.95	1.00	0.97	75
1	1.00	0.95	0.97	75
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

SVM with POLY Kernel <PES2UG23CS387>

	precision	recall	f1-score	support
0	0.85	0.95	0.89	75
1	0.94	0.83	0.88	75
accuracy			0.89	150
macro avg	0.89	0.89	0.89	150
weighted avg	0.89	0.89	0.89	150

Banknote dataset:



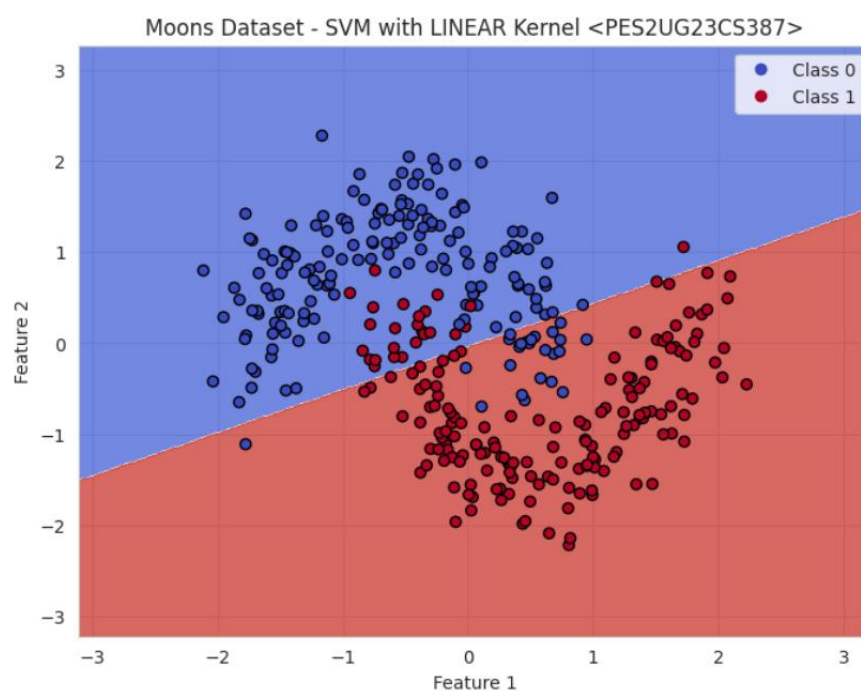
SVM with LINEAR Kernel <PES2UG23CS387>

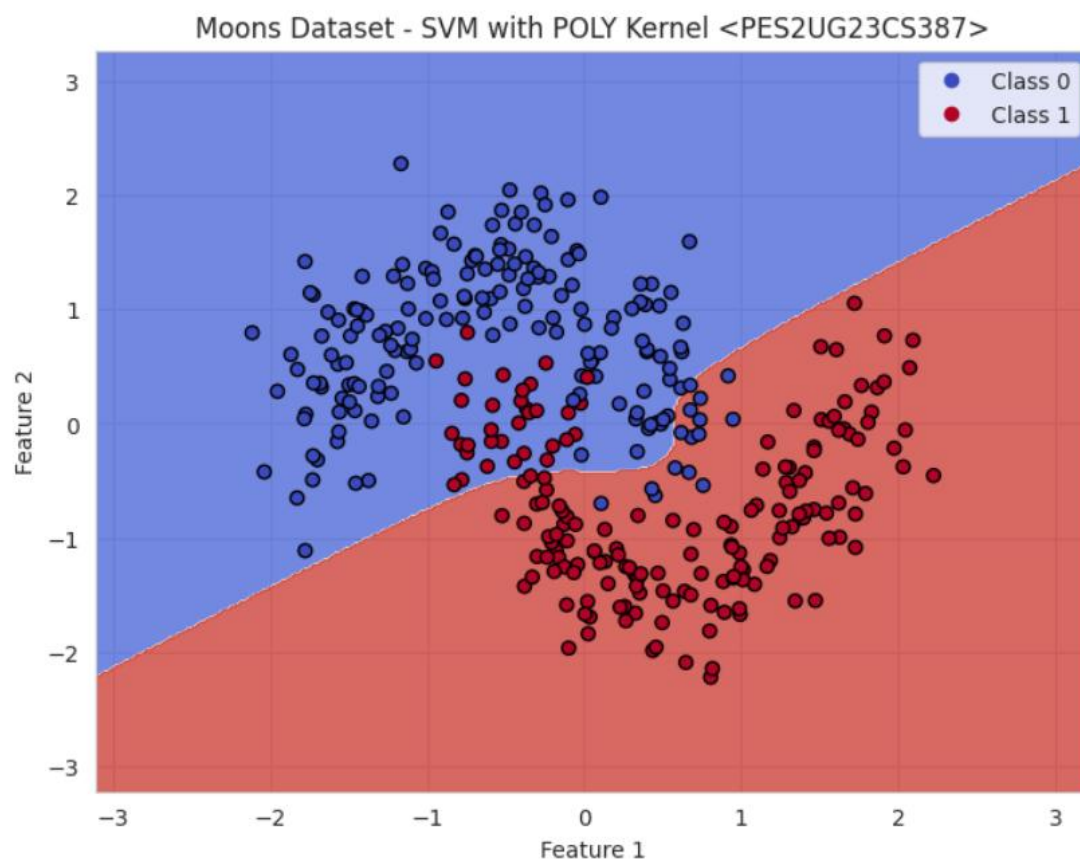
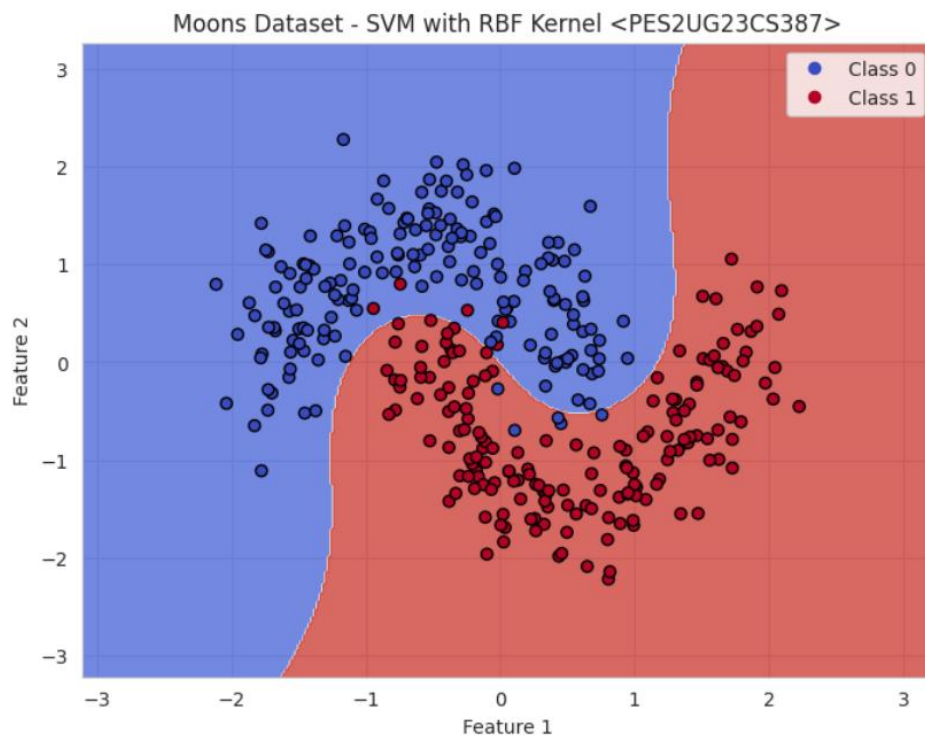
	precision	recall	f1-score	support
Forged	0.90	0.88	0.89	229
Genuine	0.86	0.88	0.87	183
accuracy			0.88	412
macro avg	0.88	0.88	0.88	412
weighted avg	0.88	0.88	0.88	412

SVM with RBF Kernel <PES2UG23CS387>				
	precision	recall	f1-score	support
Forged	0.96	0.91	0.94	229
Genuine	0.90	0.96	0.93	183
accuracy			0.93	412
macro avg	0.93	0.93	0.93	412
weighted avg	0.93	0.93	0.93	412

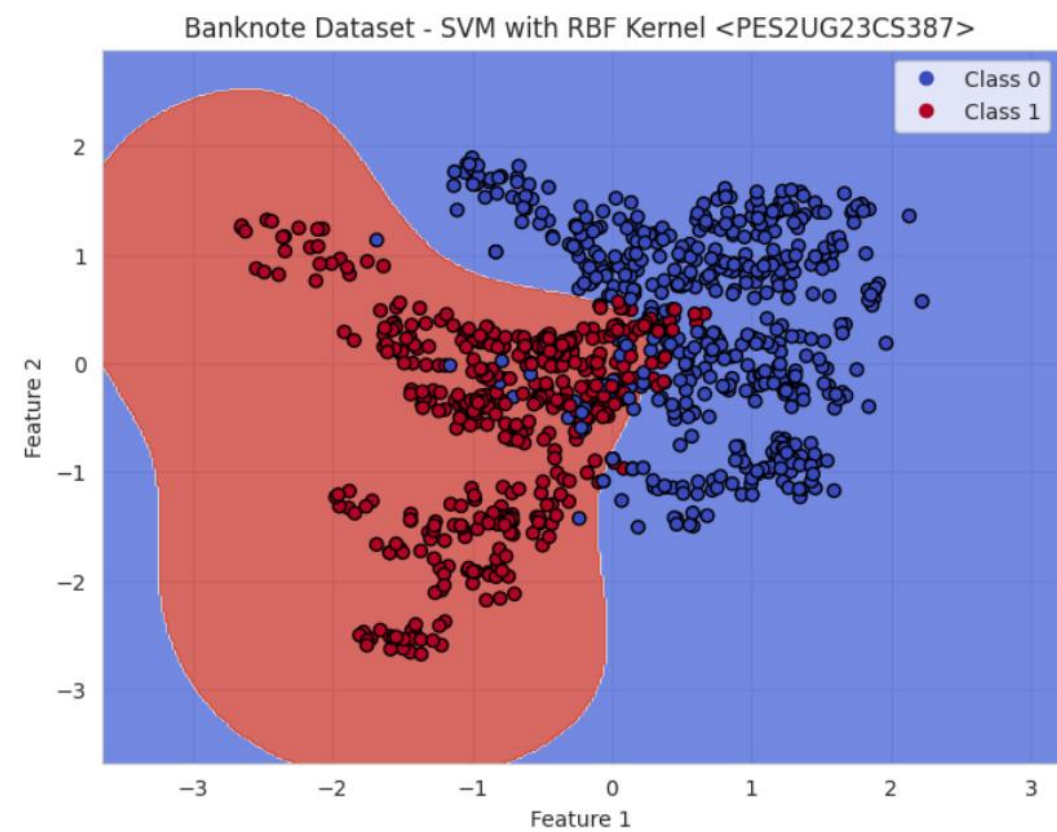
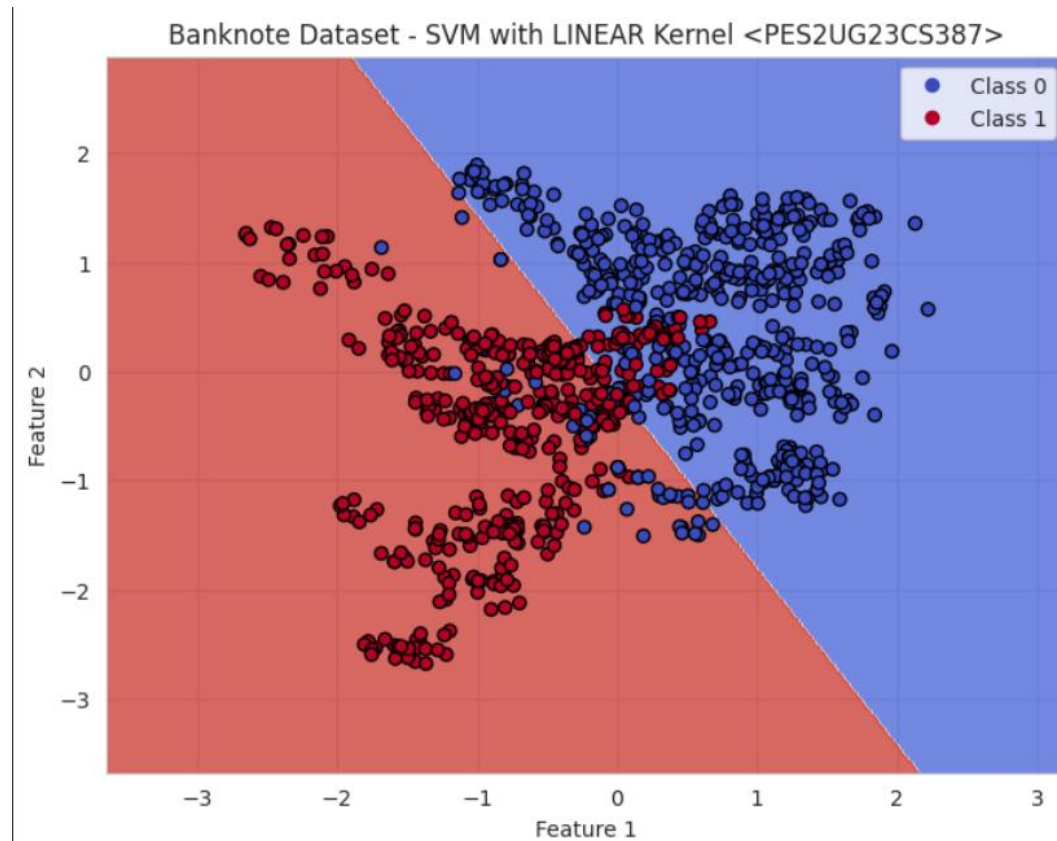
SVM with POLY Kernel <PES2UG23CS387>				
	precision	recall	f1-score	support
Forged	0.82	0.91	0.87	229
Genuine	0.87	0.75	0.81	183
accuracy			0.84	412
macro avg	0.85	0.83	0.84	412
weighted avg	0.85	0.84	0.84	412

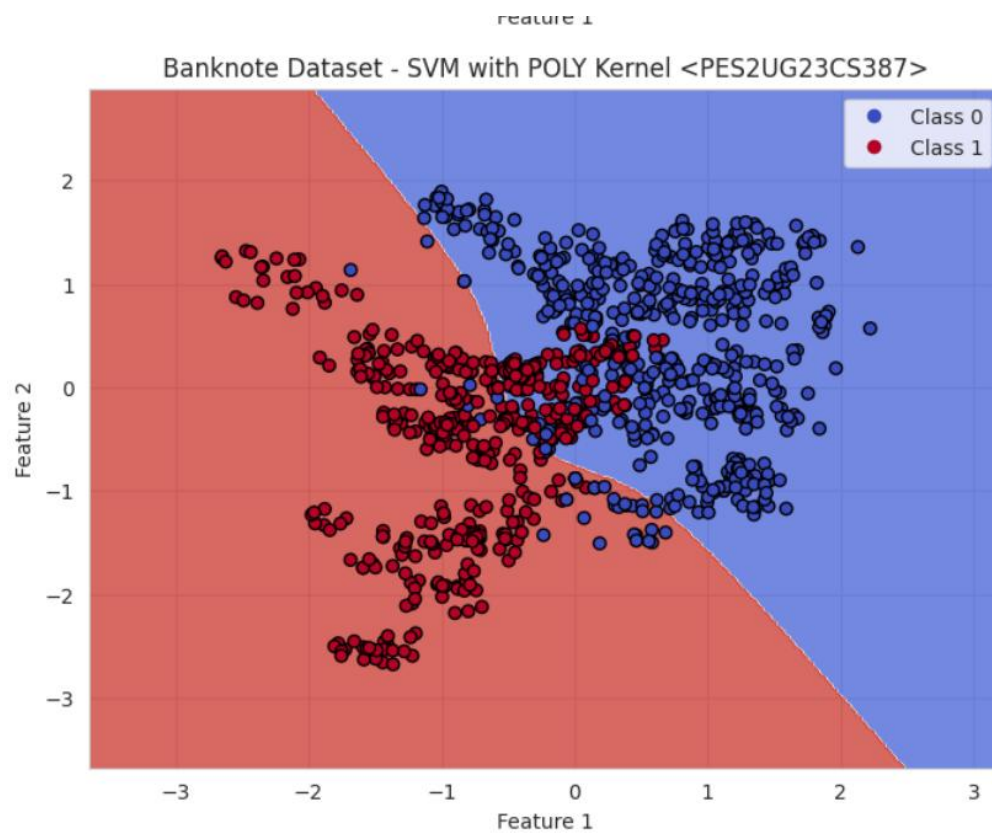
Plots: Moon





Banknote:





Margin plots:

